

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

1) Network Scenario: We consider a OneWeb-like LEO satellite communication system and focus on downlink interference management under NGSO–GSO coexistence. Each LEO satellite is equipped with 16 spot beams as illustrated in Fig. 1 [13], and serves terrestrial users within its coverage area in every time slot.

Let $\mathcal{S}(t)$ denote the set of visible LEO satellites at time slot t , and let $\mathcal{B}_s$ denote the set of beams on satellite s . We assume that every LEO satellite has the same number of beams:

$$|\mathcal{B}_s| = 16, \quad \forall s \in \mathcal{S}(t).$$

Each satellite employs a fixed beam layout with 16 spot beams arranged along the along-track (north–south) direction. For beam b on satellite s , the ground footprint at time slot t is denoted by

$$\mathcal{F}_{s,b}(t).$$

If user u lies in $\mathcal{F}_{s,b}(t)$, user u is covered by beam b on satellite s and may be considered as a candidate for subsequent user association.

The system also includes a protected GSO ground station, denoted by g . When a visible LEO satellite passes close to the GS boresight or forms a near-inline geometry with respect to the GSO direction, its downlink transmission may produce a high aggregate EPFD at the protected GS. We define the LEO satellite that poses the dominant EPFD risk to the protected GS at time slot t as the *critical satellite*:

$$s^c(t).$$

We further consider the along-track north and south neighbors of the critical satellite as candidate helper satellites for subsequent service transfer and relaying:

$$s^N(t), \quad s^S(t).$$

where $s^N(t)$ and $s^S(t)$ denote the north and south helper satellites, respectively, as shown in Fig. 2. Because the considered LEO constellation follows a regular Walker-star orbital structure, the relative geometry and beam coverage overlap among the critical satellite and its helpers are predictable. This property underpins the beam-level user reassociation and relay design in the next chapter.

2) User Association and Service Model: Let $\mathcal{U}(t)$ denote the set of users within the system service region at time slot t . Each user can be served by at most one beam on one LEO satellite in the same time slot. The user–beam association is represented by

$$x_{u,s,b}(t) = \begin{cases} 1, & \text{if user } u \text{ is served by beam } b \text{ of satellite } s, \\ 0, & \text{otherwise,} \end{cases}$$

for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, and $b \in \mathcal{B}_s$.

The set of users served by beam b on satellite s is

$$\mathcal{U}_{s,b}(t) = \{u \in \mathcal{U}(t) \mid x_{u,s,b}(t) = 1\}.$$

Hence,

$$N_{s,b}(t) = \sum_{u \in \mathcal{U}(t)} x_{u,s,b}(t). \quad (1)$$

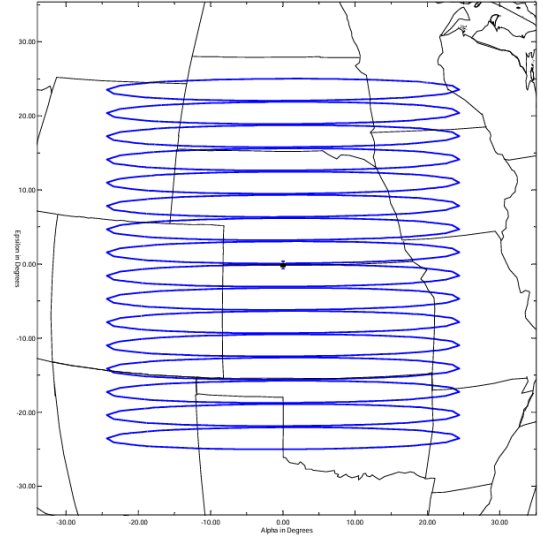


Fig. 1. Ground footprints of the OneWeb Ku-band 16-beams

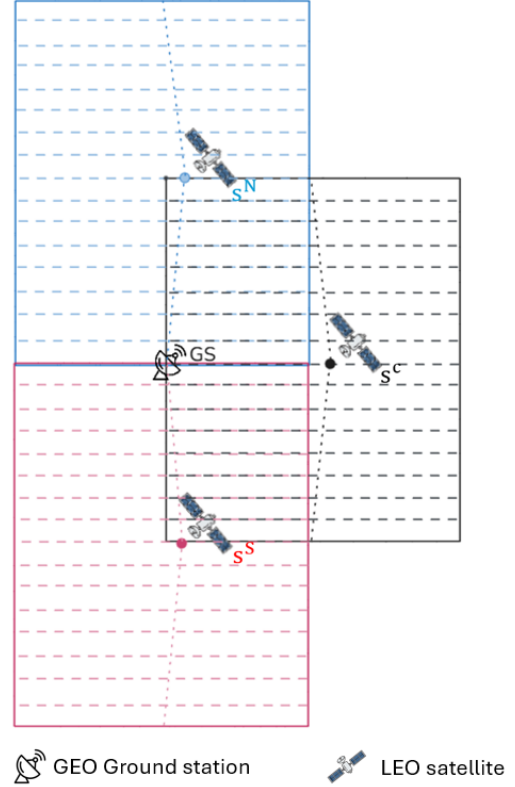


Fig. 2. Network scenario with the protected GSO ground station GS , critical satellite s^c , and north/south helper satellites s^N and s^S .

Because each user is served by only one beam, the downlink rate of a user is determined solely by its serving beam. The service quality of a user depends on the link condition, beam power, channel assignment of the serving beam, and the number of co-served users on that beam.

If user u is not within the footprint $\mathcal{F}_{s,b}(t)$ of beam s, b , that beam is not a feasible serving beam for user u . Association decisions are therefore made only among satellite–

beam pairs that can cover the user, ensuring that subsequent association, reassociation, and relay decisions respect actual beam coverage.

3) Beam Power Model: We adopt a baseline-power beam operation model. In normal operation, when LEO satellites are far from the protected GS and the EPFD violation risk is low, all active beams on each satellite are configured with the same baseline power. For beam b on satellite s , the transmit power is denoted by

$$P_{s,b}(t).$$

In normal operation,

$$P_{s,b}(t) = P^{\text{base}}, \quad \forall s \in \mathcal{S}(t), \forall b \in \mathcal{B}_s.$$

where P^{base} is the baseline power under normal conditions. The purpose of baseline power is to provide a fixed configuration that meets typical user demand without solving a full per-time-slot continuous power control problem. Because the numbers of beams and users are large, re-solving a complete power control problem in every time slot is computationally demanding and difficult to deploy in practice. We therefore use baseline-power operation as the initial configuration in normal operation and concentrate subsequent power adjustments on EPFD-risk periods involving beam shutdown and helper beam power boosting.

Each beam has a maximum transmit power $P_{s,b}^{\text{max}}$, and

$$0 \leq P_{s,b}(t) \leq P_{s,b}^{\text{max}}, \quad \forall s \in \mathcal{S}(t), \forall b \in \mathcal{B}_s. \quad (2)$$

Each satellite also has a total transmit power limit P_s^{max} :

$$\sum_{b \in \mathcal{B}_s} P_{s,b}(t) \leq P_s^{\text{max}}, \quad \forall s \in \mathcal{S}(t). \quad (3)$$

When the critical satellite $s^c(t)$ approaches the protected GS and creates an aggregate EPFD violation risk, some high-risk beams may be shut down. Let $\mathcal{B}_c^{\text{off}}(t)$ denote the set of beams switched off on the critical satellite. Then

$$P_{s^c,b}(t) = 0, \quad \forall b \in \mathcal{B}_c^{\text{off}}(t).$$

The remaining beams are *safe beams*, denoted by $\mathcal{B}_c^{\text{safe}}(t)$, and keep the baseline power:

$$P_{s^c,b}(t) = P^{\text{base}}, \quad \forall b \in \mathcal{B}_c^{\text{safe}}(t),$$

where

$$\mathcal{B}_c^{\text{safe}}(t) = \mathcal{B}_{s^c} \setminus \mathcal{B}_c^{\text{off}}(t).$$

Hence,

$$P_{s^c,b}(t) = \begin{cases} 0, & b \in \mathcal{B}_c^{\text{off}}(t), \\ P^{\text{base}}, & b \in \mathcal{B}_c^{\text{safe}}(t). \end{cases}$$

For helper satellites, beams operate at baseline power under normal conditions. When a helper must take over users from closed beams on the critical satellite, relaying helper beams may be raised up to their beam-level maximum:

$$P_{s^h,b}(t) \leq P_{s^h,b}^{\text{max}}, \quad s^h \in \{s^N(t), s^S(t)\},$$

subject to (3). In the proposed method of the next chapter, critical safe beams may absorb part of the helper traffic so that the helper satellite can release power back to its available power pool and boost the relaying helper beams. Although

transmit power may be reallocated among beams on the same helper satellite, each user is still served by only one beam, and the delivered service rate is determined only by the capacity of that serving beam.

4) SINR Model: The received quality of a user is determined by the desired signal from the serving beam, co-channel interference, and receiver noise. We adopt a OneWeb-like multi-channel frequency assignment with channel set $\mathcal{K}$ and

$$|\mathcal{K}| = 8.$$

If beam b on satellite s uses channel $k_{s,b} \in \mathcal{K}$, the user suffers co-channel interference only from beams that reuse the same channel as the serving beam.

If user u is served by beam b on satellite s , the received SINR is

$$\text{SINR}_u(t) = \frac{S_u(t)}{I_u^{\text{intra}}(t) + I_u^{\text{inter}}(t) + N_u(t)}. \quad (4)$$

where $S_u(t)$ is the desired signal power, $I_u^{\text{intra}}(t)$ and $I_u^{\text{inter}}(t)$ are intra-satellite and inter-satellite co-channel interference, respectively, and $N_u(t)$ is the receiver noise power.

Desired Signal Power: For user u served by beam b on satellite s , the desired signal power is

$$S_u(t) = x_{u,s,b}(t) \frac{P_{s,b}(t) G_{s,b \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s,u}(t)}.$$

where $P_{s,b}(t)$ is the transmit power of beam b , $G_{s,b \rightarrow u}^{\text{tx}}(t)$ is the transmit gain of beam b toward user u , G_u^{rx} is the user receive gain, and $L_{s,u}(t)$ is the total propagation loss between satellite s and user u .

The total propagation loss is modeled as

$$L_{s,u}(t) = L_{s,u}^{\text{fs}}(t) L_{s,u}^{\text{gas}}(t),$$

where $L_{s,u}^{\text{fs}}(t)$ and $L_{s,u}^{\text{gas}}(t)$ denote free-space and atmospheric gas losses, respectively. The free-space path loss is

$$L_{s,u}^{\text{fs}}(t) = \left(\frac{4\pi d_{s,u}(t)}{\lambda} \right)^2,$$

where $d_{s,u}(t)$ is the distance between satellite s and user u , and λ is the carrier wavelength.

Intra-Satellite Interference: Because different beams on the same satellite may reuse the same channel, a user may suffer interference from other co-channel beams on its serving satellite. This component is defined as intra-satellite interference.

If user u is served by beam b on satellite s , the intra-satellite interference is

$$I_u^{\text{intra}}(t) = \sum_{\substack{b' \in \mathcal{B}_s \\ b' \neq b}} \mathbf{1}(k_{s,b'} = k_{s,b}) \frac{P_{s,b'}(t) G_{s,b' \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s,u}(t)},$$

where $\mathbf{1}(k_{s,b'} = k_{s,b})$ is a channel indicator function that equals 1 if beam b' reuses the serving channel and 0 otherwise.

Inter-Satellite Interference: In addition to intra-satellite interference, a user may be interfered with by co-channel beams on other visible satellites. This component is defined as inter-satellite interference.

If user u is served by beam b on satellite s , the inter-satellite interference is

$$I_u^{\text{inter}}(t) = \sum_{\substack{s' \in \mathcal{S}(t) \\ s' \neq s}} \sum_{b' \in \mathcal{B}_{s'}} \mathbf{1}(k_{s',b'} = k_{s,b}) \frac{P_{s',b'}(t) G_{s',b' \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s',u}(t)}, \quad (8)$$

where $G_{s',b' \rightarrow u}^{\text{tx}}(t)$ is the transmit gain of interfering beam s', b' toward user u , and $L_{s',u}(t)$ is the total propagation loss between satellite s' and user u .

Under the OneWeb-like 8-channel architecture, the received quality depends on the serving beam power, antenna gain, and path loss, as well as co-channel intra-satellite and inter-satellite interference. Beam shutdown, user reassociation, and helper relaying can therefore change user-side SINR and satisfaction.

5) Capacity and User Satisfaction Model: After obtaining the user SINR, the instantaneous capacity follows from the Shannon formula. Let B_u denote the effective bandwidth of user u . The instantaneous capacity of user u at time slot t is

$$C_u(t) = B_u \log_2(1 + \text{SINR}_u(t)).$$

Because multiple users may be served by the same beam, intra-beam resource sharing is modeled by TDMA. If user u is served by beam b on satellite s , the number of served users on the beam is $N_{s,b}(t)$. The actual service rate of user u is therefore

$$R_u(t) = \frac{C_u(t)}{N_{s,b}(t)}. \quad (5)$$

where $N_{s,b}(t)$ is defined in (1). Equation (5) indicates that users on the same beam share the downlink resource of that beam through TDMA. Because each user is served by only one beam, $R_u(t)$ is determined solely by the serving beam.

Let $D_u(t)$ denote the demand rate of user u at time slot t . The satisfaction of user u is defined as

$$\text{sat}_u(t) = \min\left(\frac{R_u(t)}{D_u(t)}, 1\right). \quad (6)$$

where $\text{sat}_u(t) \in [0, 1]$. If $\text{sat}_u(t) = 1$, the demand of user u is fully met; if $0 < \text{sat}_u(t) < 1$, user u receives only partial service; if $\text{sat}_u(t) = 0$, user u receives no effective service in the time slot.

This satisfaction metric is used in the problem formulation and performance evaluation. The goal is to maximize the total satisfaction of all users subject to EPFD and power constraints.

6) Beam-Level EPFD Model: Under NGSO-GSO coexistence, the LEO satellite system must keep the aggregate EPFD at the protected GSO ground station below the regulatory limit [5], [6]. Because each LEO satellite employs 16 spot beams with different pointing directions, off-axis angles, and transmit powers toward the GS, we adopt beam-level EPFD calculation and sum the contributions in the linear domain.

For beam b on satellite s , the EPFD contribution toward the protected GS g is defined as

$$\text{EPFD}_{s,b}(t) = \frac{P_{s,b}(t) G_{s,b \rightarrow g}^{\text{tx}}(t)}{4\pi d_{s,g}^2(t)} \cdot \frac{G_g^{\text{rx}}(\theta_{g \rightarrow s}(t))}{G_{g,\text{max}}^{\text{rx}}} \cdot \frac{1}{B_{\text{ref}}},$$

where $P_{s,b}(t)$ is the beam transmit power, $G_{s,b \rightarrow g}^{\text{tx}}(t)$ is the transmit gain toward the GS, $d_{s,g}(t)$ is the satellite-GS distance, $G_g^{\text{rx}}(\theta_{g \rightarrow s}(t))$ is the GS receive gain, $G_{g,\text{max}}^{\text{rx}}$ is the

peak GS receive gain, $\theta_{g \rightarrow s}(t)$ is the off-axis angle from the GS toward satellite s , and B_{ref} is the reference bandwidth [7],

The transmit gain toward the GS is expressed as a function of the off-axis angle:

$$G_{s,b \rightarrow g}^{\text{tx}}(t) = G_{s,b}^{\text{tx}}(\phi_{s,b \rightarrow g}(t)),$$

where $\phi_{s,b \rightarrow g}(t)$ is the off-axis angle from the boresight of beam b to the protected GS. Because the 16 beams on the same satellite point in different directions, their EPFD contributions must be computed individually.

The total EPFD contribution of satellite s is the linear sum over all beams:

$$\text{EPFD}_s^{\text{sat}}(t) = \sum_{b \in \mathcal{B}_s} \text{EPFD}_{s,b}(t).$$

The aggregate EPFD from all visible LEO satellites at time slot t is

$$\text{EPFD}_{\text{agg}}(t) = \sum_{s \in \mathcal{S}(t)} \text{EPFD}_s^{\text{sat}}(t).$$

Equivalently,

$$\text{EPFD}_{\text{agg}}(t) = \sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} \text{EPFD}_{s,b}(t).$$

To ensure regulatory compliance, the system must satisfy

$$\text{EPFD}_{\text{agg}}(t) \leq \text{EPFD}_{\text{lim}}, \quad (7)$$

where EPFD_{lim} is the applicable EPFD regulatory limit. If $\text{EPFD}_{\text{agg}}(t) > \text{EPFD}_{\text{lim}}$ in a time slot, the current beam power configuration produces excessive GS interference and must be adjusted through beam shutdown or user reassociation.

B. Problem Formulation

Based on the system model above, we maximize the total user satisfaction subject to user association, beam power, satellite total power, and EPFD constraints. The joint optimization problem is formulated as follows.

Variables: The optimization variables are the user-beam association indicators $\mathbf{x}(t) = \{x_{u,s,b}(t)\}$ and the beam transmit powers $\mathbf{P}(t) = \{P_{s,b}(t)\}$.

Objective function:

$$\max_{\mathbf{x}(t), \mathbf{P}(t)} \sum_{u \in \mathcal{U}(t)} \text{sat}_u(t). \quad (8)$$

Constraints:

$$\sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} x_{u,s,b}(t) \leq 1, \quad \forall u \in \mathcal{U}(t), \quad (9)$$

$$0 \leq P_{s,b}(t) \leq P_{s,b}^{\text{max}}, \quad \forall s \in \mathcal{S}(t), \forall b \in \mathcal{B}_s. \quad (10)$$

$$\sum_{b \in \mathcal{B}_s} P_{s,b}(t) \leq P_s^{\text{max}}, \quad \forall s \in \mathcal{S}(t). \quad (11)$$

$$\text{EPFD}_{\text{agg}}(t) \leq \text{EPFD}_{\text{lim}}. \quad (12)$$

Constraint (9) ensures that each user is served by at most one satellite-beam pair in each time slot.

Constraint (10) limits each beam to $[0, P_{s,b}^{\max}]$, allowing shutdown (0) and helper boosting (up to $P_{s,b}^{\max}$).

Constraint (11) enforces the satellite total power budget $P_s^{\max}$.

Constraint (12) enforces the aggregate EPFD regulatory limit at the protected GS.

The objective in (8) maximizes the aggregate satisfaction of all users under the EPFD and power constraints. Because $\text{sat}_u(t)$ is defined in (6), the objective reflects the overall service quality in time slot t .

Because association affects TDMA sharing, SINR, service rate, and satisfaction, while beam power affects both user-side link quality and GS-side aggregate EPFD, the problem is highly coupled. Near the protected GS, beam shutdown reduces EPFD but degrades users on closed beams; reassociating them to helpers increases helper beam loading and may hurt existing helper users.

We therefore do not pursue a global optimum of this mixed-integer nonlinear program. Instead, the next chapter presents a staged heuristic framework that satisfies the EPFD constraint through safe-beam assisted power release and beam-pair local relay to improve satisfaction during critical periods.

IV. METHOD

A. Method Overview

Chapter III formulates the problem as maximizing total user satisfaction subject to user association, beam power, satellite total power, and EPFD constraints. However, the problem involves both discrete user-beam association decisions and continuous beam power configuration, and association, SINR, TDMA sharing, satisfaction, and aggregate EPFD are tightly coupled. Solving for a global optimum is therefore computationally demanding.

To reduce online decision complexity, we propose Safe-Beam Assisted Power-Release Relay (SAPR-R). SAPR-R exploits the predictability of the LEO constellation, protected GS location, GSO direction, and fixed beam layout to pre-build beam-level relation records for all time slots in the observation window. For each time slot, the system records the critical satellite, north/south helper satellites, beams on the critical satellite that must be shut down, safe beams that remain active at baseline power, and beam-pair mappings between critical and helper beams. This stage depends only on satellite geometry, beam footprints, and EPFD calculation not on user distribution and can be completed before online user reassociation.

During online operation, the system reads the beam relation record for the current time slot and performs user-aware relay decisions based on user locations and associations. First, SAPR-R applies Safe-Beam Assisted Helper Relief (SAHR) so that critical safe beams attempt to take over users on helper release beams. If reassociation does not reduce the satisfaction of the transferred user or of existing users on the safe beam, the transfer is accepted and the power released from the helper release beam is returned to the helper satellite's available power pool. Next, the helper satellite allocates the power pool to helper relay beams according to relay demand, increasing

capacity for subsequent service to closed-beam users. Finally, SAPR-R applies Beam-Pair Local Relay (BPLR): within each closed-beam/helper relay-beam pair, critical closed-beam users are ranked and allocated sequentially according to relay link quality.

In summary, the proposed method consists of the following steps:

- **Time-slot beam relation record:** Pre-record critical/helper satellites, closed/safe beams, and beam-pair mappings for each time slot.
- **Safe-Beam Assisted Helper Relief (SAHR):** Use critical safe beams to take over selected helper users and release helper satellite power without lowering satisfaction.
- **Helper relay beam boosting:** Allocate the available power pool to helper relay beams according to relay demand.
- **Beam-Pair Local Relay (BPLR):** Within each beam pair, allocate relay capacity to closed-beam users according to relay link quality.

B. Time-Slot Beam Relation Record

For each time slot t , we construct a beam relation record $\mathcal{R}_B(t)$. Figure 3 illustrates the time-slot beam status considered in this work, including the:

$$\mathcal{R}_B(t) = (s^c(t), s^N(t), s^S(t), \mathcal{B}_c^{\text{off}}(t), \mathcal{B}_c^{\text{safe}}(t), \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t), \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)).$$

This record stores the beam-level information required by SAHR and BPLR, where $s^c(t)$, $s^N(t)$, and $s^S(t)$ are the critical, north helper, and south helper satellites, respectively. $\mathcal{B}_c^{\text{off}}(t)$ is the set of beams on the critical satellite shut down due to the EPFD constraint; $\mathcal{B}_c^{\text{safe}}(t)$ is the set of safe beams that remain on at baseline power. $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$ and $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ denote the safe-beam-to-helper-release-beam mapping and the closed-beam-to-helper-relay-beam mapping, respectively.

1) Critical and Helper Satellite Identification: In each time slot t , the system identifies the critical satellite that poses the dominant EPFD risk to the protected GS from LEO-GS geometry:

$$s^c(t).$$

From simulation, EPFD violations are dominated by high-risk beams on the critical satellite; we therefore apply shutdown only to beams on $s^c(t)$, not to helper or other visible satellites. The along-track north and south neighbors of the critical satellite are selected as helpers:

$$s^N(t), \quad s^S(t).$$

2) EPFD-Driven Beam Status Determination: For each critical beam $b \in \mathcal{B}_{s^c}$, define the EPFD risk score

$$q_{c,b}(t) = \text{EPFD}_{s^c,b}(t).$$

Critical beams are sorted in descending order of $q_{c,b}(t)$, and a greedy shutdown rule is applied: at each step, shut down the beam with the highest current EPFD risk score until

$$\text{EPFD}_{\text{agg}}(t) \leq \text{EPFD}_{\text{lim}}.$$

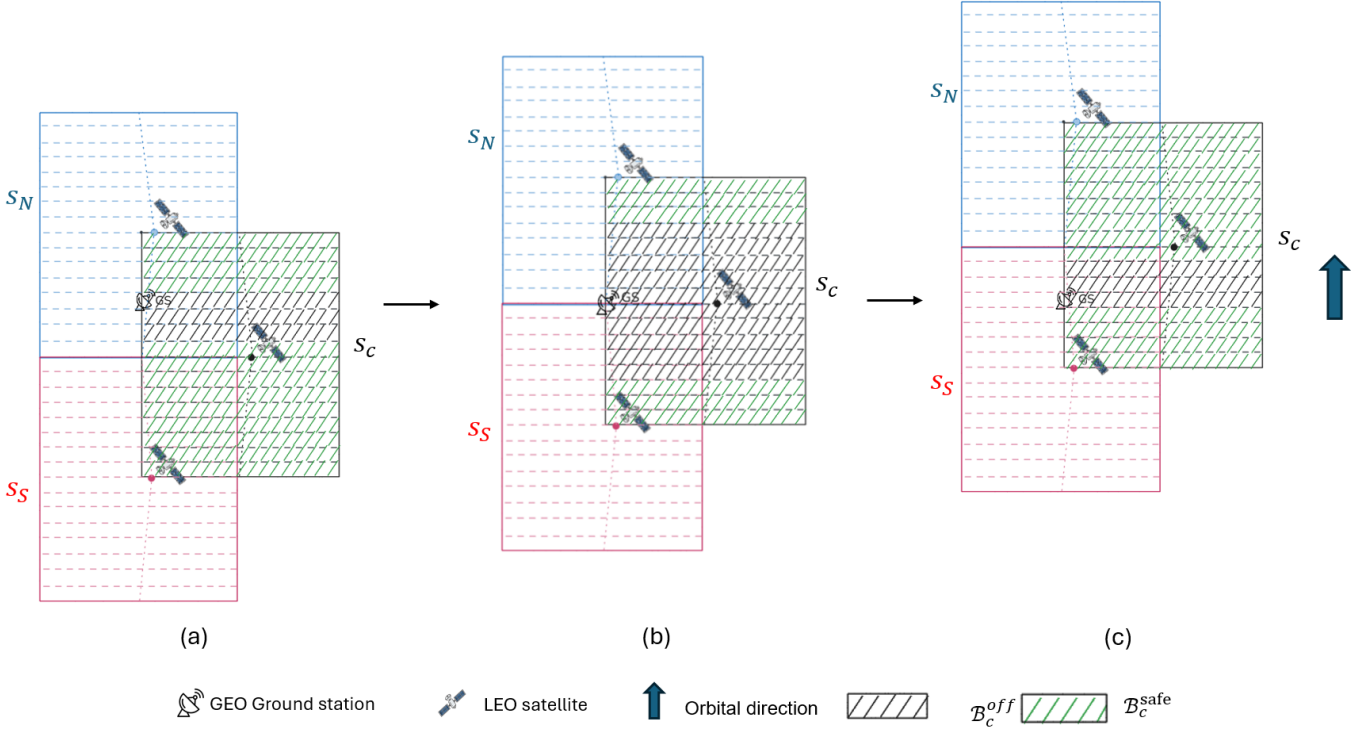


Fig. 3. Three time slots as the along-track LEO satellites (s^N , s^c , and s^S) pass the protected GSO ground station. In region S_C on the critical satellite, black-hatched beams are $\mathcal{B}_c^{\text{off}}(t)$ (closed beams) and green-hatched beams are $\mathcal{B}_c^{\text{safe}}(t)$ (safe beams). Panels (a)–(c) show earlier to later time slots.

Shut-down beams form

$$\mathcal{B}_c^{\text{off}}(t),$$

and the remaining beams form the safe set:

$$\mathcal{B}_c^{\text{safe}}(t) = \mathcal{B}_{s^c} \setminus \mathcal{B}_c^{\text{off}}(t).$$

Beam status is determined solely from beam-level EPFD contributions and the regulatory limit, without user distribution, and can be precomputed before online reassociation.

3) Beam-Pair Mapping Construction: Given $\mathcal{B}_c^{\text{off}}(t)$ and $\mathcal{B}_c^{\text{safe}}(t)$, beam-pair mappings are built from footprint overlap. For SAHR,

$$\begin{aligned} \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t) = \{ & (b_c^{\text{safe}}, b_h^{\text{rel}}) : \\ & \mathcal{F}_{s^c, b_c^{\text{safe}}}(t) \cap \mathcal{F}_{s_h, b_h^{\text{rel}}}(t) \neq \emptyset, \\ & s_h \in \{s^N(t), s^S(t)\} \}, \end{aligned}$$

which identifies which helper release-beam users a critical safe beam can take over.

For BPLR,

$$\begin{aligned} \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t) = \{ & (b_c^{\text{off}}, b_h^{\text{relay}}) : \\ & \mathcal{F}_{s^c, b_c^{\text{off}}}(t) \cap \mathcal{F}_{s_h, b_h^{\text{relay}}}(t) \neq \emptyset \}, \end{aligned}$$

which identifies which helper relay beams can serve critical closed-beam users.

Figure 4 visualizes the two beam-pair mappings stored in $\mathcal{R}_B(t)$: $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$ for SAHR and $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ for BPLR.

Under the Walker-star constellation and fixed beam layout, the critical satellite and north/south helpers exhibit near one-to-one beam overlap: north-half beams on the critical satellite

overlap mainly with south-half beams on the north helper, and south-half critical beams with north-half beams on the south helper. Relay can therefore be executed locally within beam pairs without global matching.

Helper beams are split into release beams (to free helper power) and relay beams (to serve critical closed-beam users). We assume no overlap within the same time slot:

$$\mathcal{B}_h^{\text{rel}}(t) \cap \mathcal{B}_h^{\text{relay}}(t) = \emptyset.$$

C. Safe-Beam Assisted Helper Relief

After loading $\mathcal{R}_B(t)$, the online procedure starts with SAHR. Figure 5 illustrates the SAHR process. The goal is not to serve critical closed-beam users directly, but to use critical safe beams to take over helper users so that helper satellites release power for subsequent relay boosting.

For each

$$(b_c^{\text{safe}}, b_h^{\text{rel}}) \in \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t),$$

we identify helper users currently served by b_h^{rel} that also lie in the footprint of b_c^{safe} :

$$\begin{aligned} \mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, b_h^{\text{rel}}, t) = \\ \{ v : v \in \mathcal{U}_{s_h, b_h^{\text{rel}}}(t), v \in \mathcal{F}_{s^c, b_c^{\text{safe}}}(t) \}. \end{aligned}$$

If candidate helper user v is reassociated to b_c^{safe} , the reduction in required power on the helper release beam is

$$\Delta P_v^{\text{rel}}(t) = P_{b_h^{\text{rel}}}^{\text{req, old}}(t) - P_{b_h^{\text{rel}}}^{\text{req, new}}(v, t).$$

The SAHR score is

$$S_{\text{relief}}(v, t) = \Delta P_v^{\text{rel}}(t).$$

Beam-pair mapping visualization for a given time slot t

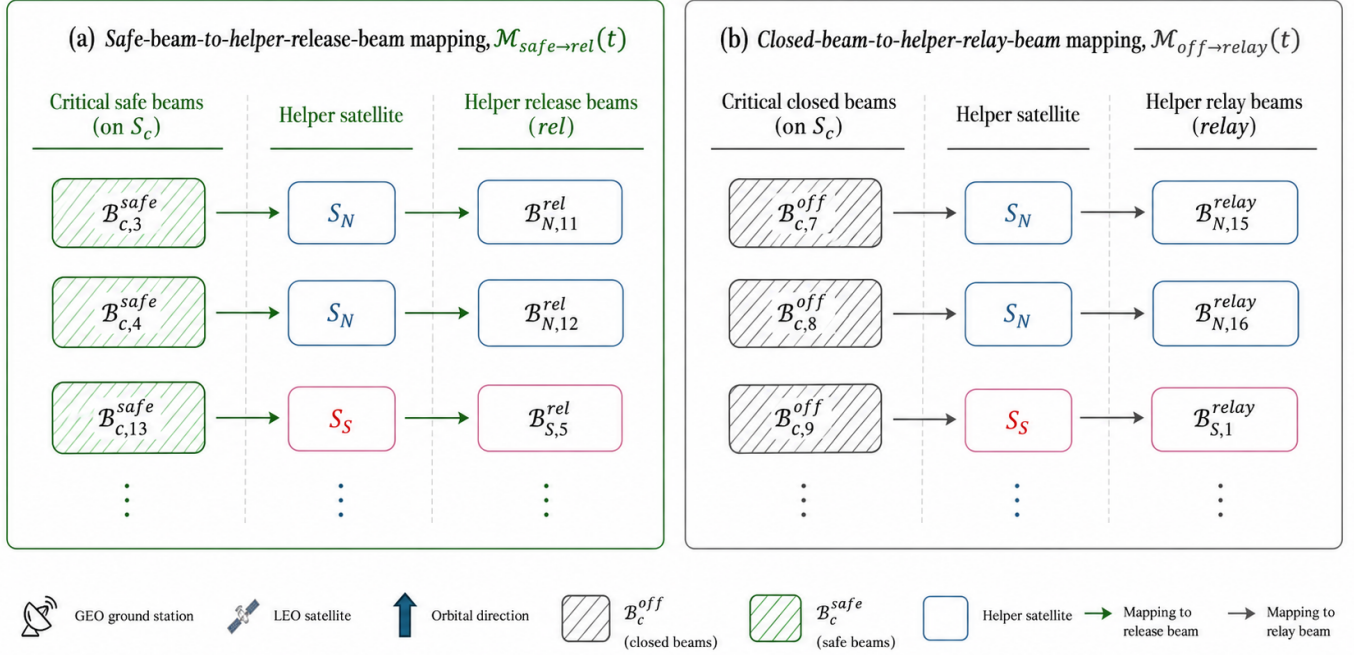


Fig. 4. Beam-pair mapping tables at time slot t . (a) $\mathcal{M}_{safe \rightarrow rel}(t)$: critical safe beams on s^c mapped to helper release beams on s^N/s^S (SAHR). (b) $\mathcal{M}_{off \rightarrow relay}(t)$: critical closed beams mapped to helper relay beams (BPLR).

Candidates are sorted in descending order of $S_{relief}(v, t)$. A trial reassociation is accepted only if the transferred user does not lose satisfaction and no original user on the safe beam loses satisfaction:

$$\begin{aligned} \text{sat}_v^{safe}(t) &\geq \text{sat}_v^{old}(t), \\ \text{sat}_w^{new}(t) &\geq \text{sat}_w^{old}(t), \quad \forall w \in \mathcal{U}_{b_{safe}^{ori}}(t). \end{aligned}$$

Accepted transfers return $\Delta P_v^{rel}(t)$ to the helper power pool. For helper satellite s_h , total released power is

$$P_{s_h}^{rel}(t) = \sum_{v \in \mathcal{A}_{rel}(s_h, t)} \Delta P_v^{rel}(t),$$

where $\mathcal{A}_{rel}(s_h, t)$ is the set of helper users successfully taken over by safe beams.

Unused power under baseline operation is

$$P_{s_h}^{unused}(t) = P_{s_h}^{max} - \sum_{b \in \mathcal{B}_{s_h}} P^{base},$$

so the relay boosting pool is

$$P_{s_h}^{pool}(t) = P_{s_h}^{unused}(t) + P_{s_h}^{rel}(t).$$

All helper beams except release and relay beams remain at baseline power P^{base} .

D. Helper Relay Beam Boosting

After SAHR, helper satellite s_h uses $P_{s_h}^{pool}(t)$ to boost helper relay beams. The relay beam set is

$$\begin{aligned} \mathcal{B}_{s_h}^{relay}(t) &= \{b_h^{relay} : \exists b_c^{off} \in \mathcal{B}_c^{off}(t) : \\ &\quad (b_c^{off}, b_h^{relay}) \in \mathcal{M}_{off \rightarrow relay}(t)\}. \end{aligned}$$

Extra power needed to raise a relay beam from baseline to its upper bound is

$$\Delta P_{b_h^{relay}}^{max}(t) = P_{s_h, b_h^{relay}}^{max} - P^{base}.$$

Relay demand for each helper relay beam is

$$D_{b_h^{relay}}^{relay}(t) = |\mathcal{U}_{cand}(b_h^{relay}, t)|,$$

where $\mathcal{U}_{cand}(b_h^{relay}, t)$ is the set of critical users in the mapped closed beam that can be covered by b_h^{relay} .

If $P_{s_h}^{pool}(t)$ suffices to boost all relay beams to their upper bounds, every relay beam is fully boosted. Otherwise, we apply relay-demand-priority boosting: relay beams are sorted by $D_{b_h^{relay}}^{relay}(t)$ in descending order, and higher-demand beams receive power first.

If the remaining pool can raise the current beam to $P_{s_h, b_h^{relay}}^{max}$, it is set to the upper bound. If the pool cannot fully boost the next beam but is still positive, the remainder is assigned as a partial boost above P^{base} , avoiding wasted pool power and increasing available relay capacity.

All helper relay beams then proceed to BPLR with final transmit powers that determine their relay capacities.

E. Beam-Pair Local Relay

After relay-beam boosting, BPLR runs within each closed-beam-to-helper-relay-beam pair. Figure 6 illustrates the BPLR process. For

$$(b_c^{off}, b_h^{relay}) \in \mathcal{M}_{off \rightarrow relay}(t),$$

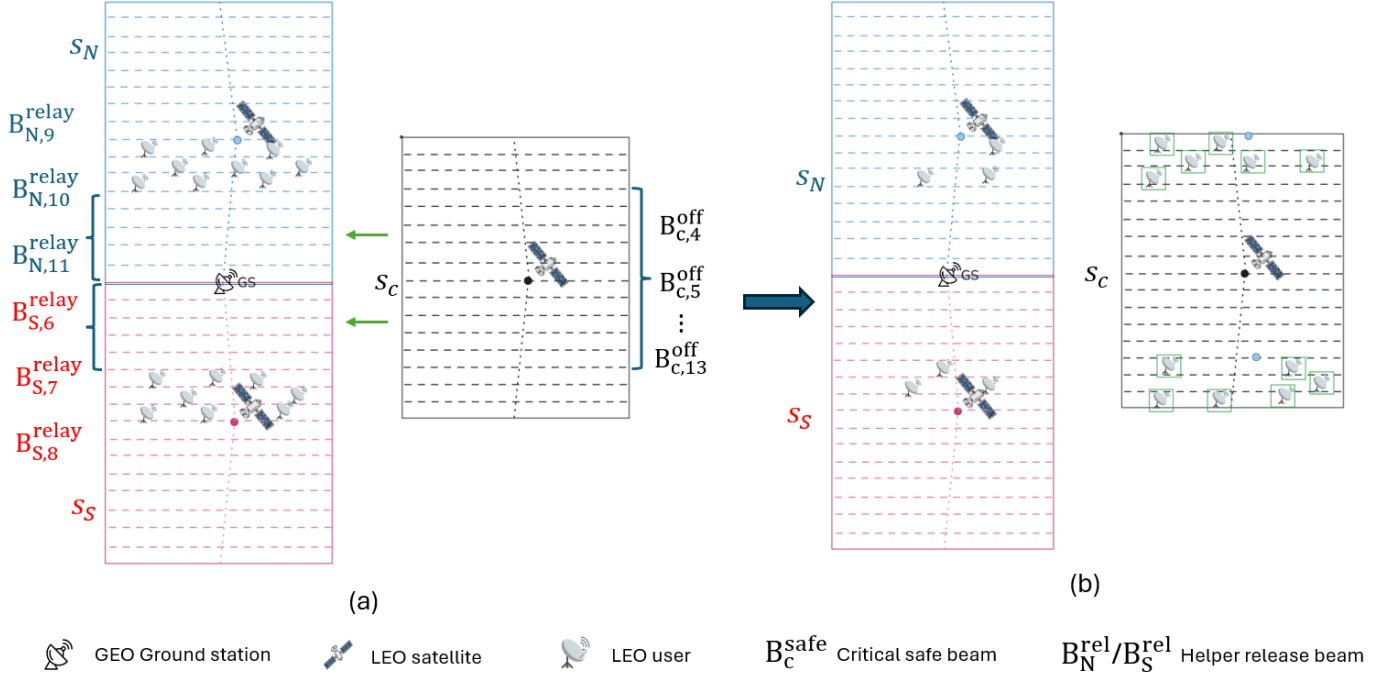


Fig. 5. Illustration of Safe-Beam Assisted Helper Relief (SAHR). Critical safe beams take over eligible helper users from helper release beams, and the released power is returned to the helper satellite power pool.

we identify critical users served by b_c^{off} that also lie in the footprint of b_h^{relay} :

$$\mathcal{U}_{\text{pair}}(b_c^{\text{off}}, b_h^{\text{relay}}, t) = \left\{ u : u \in \mathcal{U}_{s^c, b_c^{\text{off}}}(t), u \in \mathcal{F}_{s_h, b_h^{\text{relay}}}(t) \right\}.$$

The expected achievable rate if candidate critical user u is served by the helper relay beam is

$$R_{u, b_h^{\text{relay}}}(t).$$

The BPLR score is

$$S_{\text{relay}}(u, t) = R_{u, b_h^{\text{relay}}}(t).$$

Candidates are sorted in descending order of $S_{\text{relay}}(u, t)$. The helper relay beam's remaining relay capacity (after protecting existing helper users) is computed from its final transmit power.

Remaining capacity is allocated sequentially to sorted users. If capacity covers demand $D_u(t)$, u is reassociated to the helper relay beam and satisfaction is updated. If the next user cannot be fully served but capacity remains positive, partial capacity is assigned and the pair process stops. When capacity is exhausted, the pair relay process ends.

All relay decisions respect the single-association constraint: a relayed user moves from the critical closed beam to the helper relay beam and is served by only one beam in the time slot.

F. Overall Algorithm

The online procedure uses a single time-slot-based algorithm. In each time slot, SAPR-R first runs SAHR and relay-beam boosting, then runs BPLR using the updated helper relay-beam powers. The complete procedure is given in Algorithm 1.

Algorithm 1 SAPR-R Online Operation

Input: $\{\mathcal{R}_B(t)\}_{t=1}^T$, initial $\mathbf{x}(t)$, $\mathcal{U}(t)$, $D_u(t)$

Input: P^{base} , $P_{s,b}^{\text{max}}$, P_s^{max} , $F_{s,b}(t)$, capacity/satisfaction model

Output: $\mathbf{x}^{\text{final}}(t)$, $P_{s,b}^{\text{final}}(t)$, $\text{sat}_u^{\text{final}}(t)$

```

1: for  $t = 1$  to  $T$  do
2:   Load  $\mathcal{R}_B(t)$ ; initialize beam powers per record and baseline operation
3:   Stage 1: SAHR and relay-beam boosting
4:   for all  $(b_c^{\text{safe}}, b_h^{\text{rel}}) \in \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$  do
5:     Build candidate helper users; compute  $\Delta P_v^{\text{rel}}(t)$ 
6:     Sort candidates by  $\Delta P_v^{\text{rel}}(t)$  descending
7:     for all  $v$  in sorted order do
8:       Trial reassociation  $v: b_h^{\text{rel}} \rightarrow b_c^{\text{safe}}$ 
9:       if  $v$  not degraded and users on  $b_c^{\text{safe}}$  not degraded then
10:        Accept; update association and  $P_{s_h}^{\text{pool}}(t)$ 
11:      end if
12:    end for
13:   end for
14:   for all  $s_h \in \{s^N(t), s^S(t)\}$  do
15:     Extract  $\mathcal{B}_{s_h}^{\text{relay}}(t)$ ; compute relay demand; sort by demand
16:     for all  $b_h^{\text{relay}} \in \mathcal{B}_{s_h}^{\text{relay}}(t)$  in sorted order do
17:       Allocate  $P_{s_h}^{\text{pool}}(t)$  (full boost, partial boost, or baseline)
18:     end for
19:     Check  $\sum_b P_{s,b}(t) \leq P_s^{\text{max}}$ 
20:   end for
21:   Stage 2: Beam-pair local relay
22:   for all  $(b_c^{\text{off}}, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$  do
23:     Build candidate critical users; sort by relay link quality
24:     Compute remaining relay capacity of  $b_h^{\text{relay}}$  (protect helper users)
25:     for all  $u$  in sorted order do
26:       if capacity  $\geq D_u(t)$  then
27:         Reassociate  $u$  to  $b_h^{\text{relay}}$ ; update  $\text{sat}_u(t)$  and capacity
28:       else if remaining capacity  $> 0$  then
29:         Partial allocation to  $u$ ; set remaining capacity to 0; break
30:       else
31:         break
32:       end if
33:     end for
34:   end for
35:   Validate single association, beam power, satellite total power, and EPFD constraints
36:   Store  $\mathbf{x}^{\text{final}}(t)$ ,  $P_{s,b}^{\text{final}}(t)$ ,  $\text{sat}_u^{\text{final}}(t)$ 
37: end for

```

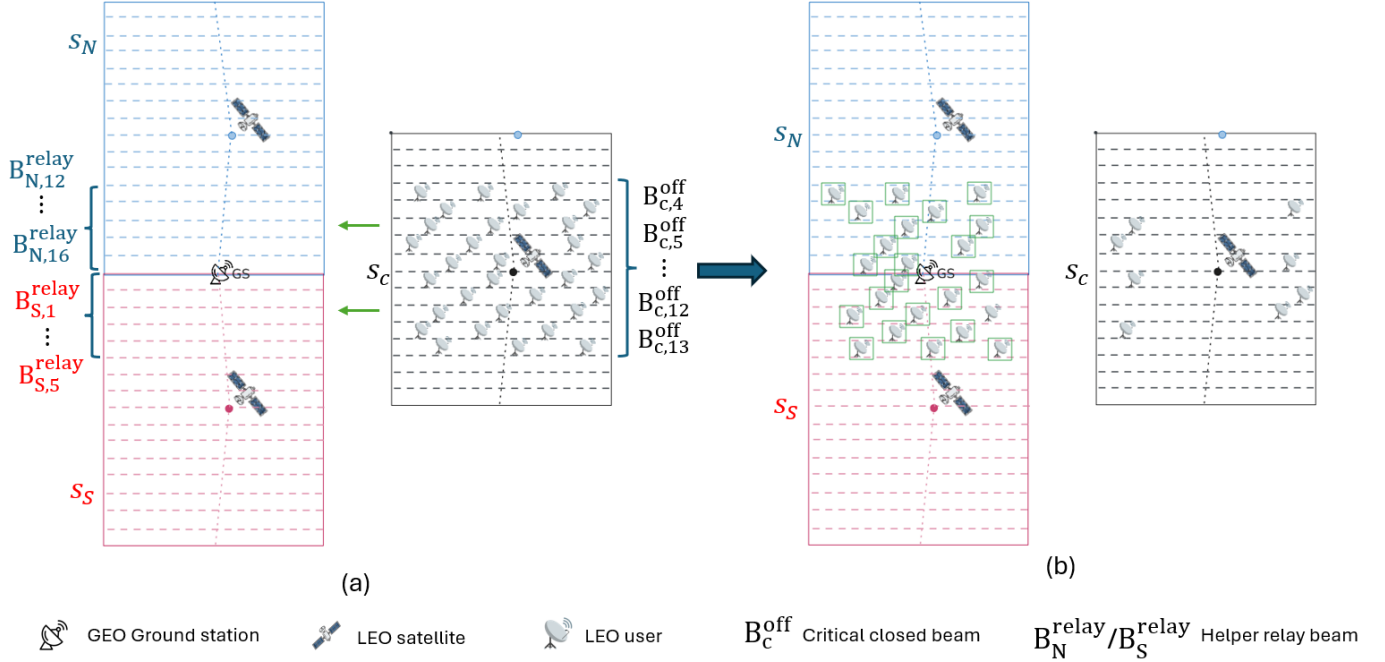


Fig. 6. Illustration of Beam-Pair Local Relay (BPLR). Candidate critical users in closed beams are reassigned to the corresponding helper relay beams according to $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$.

G. Complexity Discussion

The proposed method avoids solving the mixed-integer nonlinear program directly. Because $\mathcal{R}_B(t)$ is precomputed, the online procedure does not re-search critical/helpers or mappings; it performs local reassociation and relay only within predefined beam pairs.

Let K_t^{rel} and K_t^{relay} denote the numbers of SAHR helper candidates and BPLR critical candidates in time slot t . In Algorithm 1, building candidates and computing released power are linear in K_t^{rel} , and sorting dominates:

$$\mathcal{O}(K_t^{\text{rel}} \log K_t^{\text{rel}}).$$

In the BPLR stage, sorting by relay link quality and sequential allocation yield

$$\mathcal{O}(K_t^{\text{relay}} \log K_t^{\text{relay}}).$$

Over all time slots,

$$\mathcal{O}\left(\sum_{t=1}^T K_t^{\text{rel}} \log K_t^{\text{rel}} + K_t^{\text{relay}} \log K_t^{\text{relay}}\right).$$

Online decision is thus limited to local sorting and sequential allocation within precomputed beam pairs, rather than global association and continuous power optimization every time slot.

H. Summary

This chapter presented SAPR-R to address the tradeoff between EPFD compliance and user satisfaction when the critical satellite approaches the protected GS. We first build time-slot beam relation records from predictable geometry and footprint overlap, then run SAHR, relay-beam boosting, and BPLR online.

The staged design decouples user association and beam power allocation into local steps: critical safe beams release helper power; helper relay beams receive extra transmit power by relay demand; and closed-beam users are served within beam pairs. The method improves total user satisfaction during critical periods while satisfying the EPFD constraint.

V. EVALUATION

A. Evaluation Overview

This chapter evaluates the performance of the proposed Safe-Beam Assisted Power-Release Relay (SAPR-R) method, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical satellite pass over the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical pass, and further supports the rationale of designing a relay-based recovery method. Second, we compare different interference mitigation methods in terms of service performance and relay capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of SAPR-R under different interference constraint conditions.

B. Simulation Setup

To evaluate the performance of the proposed SAPR-R method under the NGSO–GSO coexistence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further

TABLE I
SIMULATION PARAMETERS.

evaluate the performance of different interference mitigation methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at each time slot. Based on the system information generated by STK, the dynamic geometric variation during the critical satellite pass near the GS can be simulated.
- **MATLAB:** MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam–user association, critical/helper satellite relationship, and critical-beam/helper-beam mappings. MATLAB is then used to execute the proposed SAPR-R method and calculate the EPFD and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table I. The table integrates the settings related to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

C. Compared Schemes

D. Evaluation Results

- 1) EPFD Compliance and Beam Shutdown:
- 2) User Satisfaction under Different User Loads:
- 3) Relay Capability and Satisfaction Distribution:
- 4) Impact of Different EPFD Limits:

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